Conjecture 00000001546: a polynomial growth obstruction

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1 Two incompatible necessary clauses

Both source languages assert that the maximum bisector-line count has upper growth $ne^{O(\sqrt{\log n})}$ and that no polynomial upper bound exists. These clauses are incompatible for *any* candidate counting function. We use the ordinary meaning of a polynomial upper bound: $O(n^d)$ for some fixed nonnegative integer d. In fact, the asserted upper growth implies $O(n^2)$.

This does not assign a value to a geometric counting function or replace bisector lines by an artificial model. It refutes the necessary numerical conjunction regardless of what the actual count is. We do not determine the correct maximal halving-line count, resolve variants of the source's geometric definition, or deny the compatible lower growth $ne^{\Omega(\sqrt{\log n})}$.

2 An explicit analytic bound

Let $x \geq 1$ and $C \in \mathbb{R}$. Set $u = \sqrt{\log x}$. Since $(u - C/2)^2 \geq 0$ and $u^2 = \log x$,

$$C\sqrt{\log x} \leq \log x + \frac{C^2}{4}.$$

Exponentiating and multiplying by $x \geq 0$ gives the exact bound

$$xe^{C\sqrt{\log x}} \leq e^{C^2/4}x^2. \tag{1}$$

This holds for every real C, with an explicit constant and the fixed cutoff $x \geq 1$. Hence every function that is $O(ne^{C\sqrt{\log n}})$ for some C is $O(n^2)$.

The proof also treats the O *inside* the exponential directly. If $h(n) = O(\sqrt{\log n})$, its genuine eventual norm bound gives $h(n) \leq C\sqrt{\log n}$ beyond a cutoff. Apply (1) there to obtain

$$ne^{h(n)} = O(n^2).$$

Any eventually equal function has the same bound. For a one-sided counting bound $f(n) \leq ne^{C\sqrt{\log n}}$, Lean explicitly requires $f(n) \geq 0$; the natural-valued specialization supplies this automatically. No negative-function loophole is used.

3 Formal conclusion and verification

Mathlib's actual `Asymptotics.IsBigO` at `atTop` encodes all bound constants and cutoffs. The final statements quantify over every candidate real-valued function, and separately every natural-valued count, and deny the conjunction of the printed upper-growth condition with absence of every $O(n^d)$. The witness polynomial exponent is the actual integer $d = 2$. Real logarithm, square root, exponential, eventual equality and Big-O transitivity are all proved with their actual analytic meanings.

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. A new root build, all eleven theorem axiom audits and full official `leanchecker --fresh Main` replay are included. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted, with a separate supervising Codex agent reviewing source, proof, report and rendered PDF. This is AI review, not human certification. Organizer acceptance is pending.